Supplement to “E-Bayesian and H-Bayesian estimation of a single server Markovain queue with balking”

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Abstract

This supplementary material presents the complete numerical results supporting the proposed estimation methods for the traffic intensity (ρ) and the average queue length (L_q) in the considered queueing model. The results include classical estimators, namely the Maximum Likelihood Estimator (MLE) and Uniformly Minimum Variance Unbiased Estimator (UMVUE), together with Bayesian, E-Bayesian, and Hierarchical Bayesian (H-Bayesian) estimators under a specified prior using the Squared Error Loss Function (SELF) and Precautionary Loss Function (PLF). Bayesian and H-Bayesian estimation based on Markov Chain Monte Carlo (MCMC) under a Gamma prior are also reported. In addition, predictive probabilities obtained using both analytical expressions and the Sampling Importance Resampling (SIR) method are provided for the Bayesian and H-Bayesian approaches. The supplementary material further includes extensive simulation results and the analysis of a real-life dataset to demonstrate the practical applicability of the proposed methodologies.

Table 1: Mean estimates and MSE of ρ using MLE

n	10	20	50	100	150	200
ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.1	0.10190 (0.01113)	0.10105 (0.00487)	0.09980 (0.00200)	0.09865 (0.00100)	0.09963 (0.00064)	0.09998 (0.00054)
0.2	0.19750 (0.01835)	0.20545 (0.01065)	0.20186 (0.00413)	0.19792 (0.00191)	0.20085 (0.00131)	0.19933 (0.00096)
0.5	0.49560 (0.04636)	0.49870 (0.02519)	0.49964 (0.00970)	0.49840 (0.00462)	0.49936 (0.00337)	0.50104 (0.00248)
0.7	0.69450 (0.07221)	0.70660 (0.03248)	0.69606 (0.01462)	0.70123 (0.00636)	0.69994 (0.00474)	0.70242 (0.00343)
0.9	0.91700 (0.09188)	0.89270 (0.04563)	0.90910 (0.01868)	0.90189 (0.00879)	0.90182 (0.00569)	0.89936 (0.00454)
0.99	0.99960 (0.10303)	0.98430 (0.04802)	0.99060 (0.01962)	0.98848 (0.01047)	0.99328 (0.00681)	0.98918 (0.00471)

Table 2: Mean estimates and MSE of ρ using UMVUE

n	10	20	50	100	150	200
ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.1	0.10222 (0.01062)	0.09985 (0.00498)	0.09812 (0.00213)	0.09954 (0.00094)	0.09914 (0.00063)	0.10015 (0.00050)
0.2	0.19970 (0.01903)	0.19580 (0.00964)	0.20148 (0.00400)	0.19935 (0.00188)	0.20027 (0.00132)	0.19994 (0.00102)
0.5	0.49720 (0.04638)	0.49680 (0.02498)	0.50520 (0.01060)	0.50262 (0.00480)	0.49940 (0.00321)	0.50026 (0.00256)
0.7	0.70380 (0.07336)	0.70330 (0.03541)	0.69284 (0.01466)	0.69819 (0.00683)	0.69810 (0.00479)	0.69875 (0.00364)
0.9	0.91970 (0.09507)	0.89195 (0.04294)	0.90296 (0.01809)	0.90092 (0.009194)	0.89506 (0.00606)	0.89921 (0.00428)
0.99	0.98420 (0.10028)	0.98155 (0.05019)	0.99470 (0.01924)	0.98874 (0.00990)	0.99002 (0.00630)	0.98830 (0.00463)

Table 3: Mean estimates and MSE of ρ for Bayesian estimation under gamma prior ($\alpha = 2$, $\beta = 3$)

	n	10	20	50	100	150	200
Loss function	ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.1	0.22845 (0.01723)	0.17234 (0.00749)	0.13058 (0.00246)	0.11718 (0.00112)	0.11064 (0.00072)	0.10832 (0.00053)
	0.2	0.30396 (0.02195)	0.25979 (0.01125)	0.22569 (0.00425)	0.21143 (0.00205)	0.20917 (0.00137)	0.20786 (0.00102)
	0.5	0.51275 (0.02928)	0.51087 (0.01964)	0.50632 (0.00944)	0.50572 (0.00491)	0.50237 (0.00328)	0.50213 (0.00247)
	0.7	0.61254 (0.02834)	0.65852 (0.01973)	0.68731 (0.01106)	0.70038 (0.00650)	0.69790 (0.00450)	0.69815 (0.00343)
	0.9	0.70885 (0.02436)	0.76629 (0.01619)	0.82922 (0.00855)	0.86096 (0.00513)	0.87556 (0.00382)	0.88068 (0.00307)
	0.99	0.73517 (0.02267)	0.79859 (0.01441)	0.86994 (0.00673)	0.90591 (0.00368)	0.92254 (0.00255)	0.93313 (0.00193)
PLF	0.1	0.26344 (0.06997)	0.19274 (0.04079)	0.13967 (0.01817)	0.12193 (0.00950)	0.11386 (0.00644)	0.11075 (0.00487)
	0.2	0.33854 (0.06918)	0.28055 (0.04152)	0.23493 (0.01846)	0.21622 (0.00960)	0.21241 (0.00648)	0.21031 (0.00490)
	0.5	0.54191 (0.05831)	0.53021 (0.03867)	0.51558 (0.01853)	0.51054 (0.00966)	0.50563 (0.00651)	0.50458 (0.00491)
	0.7	0.63670 (0.04830)	0.67404 (0.03105)	0.69548 (0.01634)	0.70503 (0.00930)	0.70113 (0.00645)	0.70060 (0.00490)
	0.9	0.72699 (0.03629)	0.77733 (0.02209)	0.83453 (0.01062)	0.86400 (0.00609)	0.87777 (0.00443)	0.88246 (0.00355)
	0.99	0.75147 (0.03260)	0.80801 (0.01884)	0.87394 (0.00798)	0.90798 (0.00416)	0.92395 (0.00282)	0.93419 (0.00210)

Table 4: Mean estimates and MSE of ρ for E-Bayesian estimation with $c = 1.5$

	n	10	20	50	100	150	200
Loss function	ρ	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)
SELF	0.1	* (*)	* (*)	* (*)	0.10408 (0.00103)	0.10294 (0.00068)	0.10292 (0.00051)
	0.2	* (*)	* (*)	0.20788 (0.00410)	0.20005 (0.00198)	0.20228 (0.00134)	0.20198 (0.00101)
	0.5	* (*)	0.50014 (0.02058)	0.50387 (0.00980)	0.50231 (0.004984)	0.49936 (0.00331)	0.49939 (0.00249)
	0.7	0.59876 (0.03038)	0.64917 (0.02103)	0.68646 (0.01146)	0.69902 (0.00658)	0.69886 (0.00458)	0.70264 (0.00348)
	0.9	0.70452 (0.02586)	0.76674 (0.01655)	0.82852 (0.00872)	0.85759 (0.00528)	0.87557 (0.00382)	0.88065 (0.00311)
	0.99	0.73170 (0.02403)	0.79989 (0.01473)	0.86957 (0.00688)	0.90716 (0.00366)	0.92151 (0.00260)	0.93196 (0.00198)
PLF	0.1	* (*)	* (*)	* (*)	0.10892 (0.00968)	0.10620 (0.00652)	0.10538 (0.00492)
	0.2	* (*)	* (*)	0.21748 (0.01921)	0.20495 (0.00979)	0.20557 (0.00657)	0.20446 (0.00495)
	0.5	* (*)	0.52101 (0.04173)	0.51353 (0.01931)	0.50725 (0.00987)	0.50266 (0.00661)	0.50187 (0.00497)
	0.7	0.62584 (0.05415)	0.66602 (0.03369)	0.69495 (0.01697)	0.70375 (0.00945)	0.70214 (0.00654)	0.70311 (0.00495)
	0.9	0.72424 (0.03944)	0.77815 (0.02283)	0.83395 (0.01087)	0.86073 (0.00629)	0.87778 (0.00444)	0.88244 (0.00357)
	0.99	0.74935 (0.03530)	0.80958 (0.01938)	0.87366 (0.00817)	0.90923 (0.00413)	0.92294 (0.00288)	0.93304 (0.00216)

* These values are not available

Table 5: Mean estimates and MSE of ρ for H-Bayesian estimation with $c = 1.5$

	n	10	20	50	100	150	200
Loss function	ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.1	0.12408 (0.01148)	0.11816 (0.00579)	0.10929 (0.00217)	0.10374 (0.00103)	0.10295 (0.00068)	0.10359 (0.00052)
	0.2	0.22573 (0.01971)	0.21302 (0.01032)	0.20599 (0.00407)	0.20206 (0.00201)	0.20227 (0.00134)	0.20254 (0.00101)
	0.5	0.47497 (0.03133)	0.49682 (0.02064)	0.50681 (0.00988)	0.50213 (0.00499)	0.50390 (0.00334)	0.50005 (0.00249)
	0.7	0.60907 (0.03051)	0.65007 (0.02078)	0.68776 (0.01137)	0.69775 (0.00661)	0.69672 (0.00456)	0.70244 (0.00348)
	0.9	0.70723 (0.02588)	0.76772 (0.01666)	0.82867 (0.00868)	0.86194 (0.00518)	0.87269 (0.00384)	0.88419 (0.00309)
	0.99	0.74070 (0.02394)	0.81111 (0.01402)	0.87342 (0.00663)	0.90709 (0.00367)	0.92407 (0.00251)	0.93315 (0.00194)
PLF	0.1	0.15132 (0.05449)	0.13766 (0.03900)	0.11870 (0.018823)	0.10860 (0.00972)	0.10622 (0.00654)	0.10606 (0.00493)
	0.2	0.26135 (0.07126)	0.23536 (0.04470)	0.21563 (0.01929)	0.20697 (0.00982)	0.20556 (0.00659)	0.20502 (0.00496)
	0.5	0.50881 (0.06768)	0.51788 (0.04212)	0.51650 (0.01936)	0.50707 (0.00988)	0.50721 (0.00662)	0.50254 (0.00497)
	0.7	0.63573 (0.05332)	0.66680 (0.03346)	0.69619 (0.01685)	0.70251 (0.00951)	0.69999 (0.00655)	0.70492 (0.00497)
	0.9	0.72689 (0.03932)	0.77916 (0.02287)	0.83409 (0.01083)	0.86501 (0.00614)	0.87493 (0.00448)	0.88596 (0.00351)
	0.99	0.75787 (0.03434)	0.82018 (0.01815)	0.87736 (0.00786)	0.90916 (0.00414)	0.92545 (0.00276)	0.93421 (0.00211)

Table 6: Mean estimates and MSE of L_q using MLE

n	10	20	50	100	150	200
L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.15	0.16854 (0.01341)	0.15926 (0.00671)	0.15469 (0.00244)	0.15287 (0.00128)	0.15026 (0.00083)	0.15036 (0.00060)
0.20	0.21190 (0.01888)	0.20519 (0.00868)	0.20470 (0.00358)	0.20092 (0.00179)	0.20241 (0.00117)	0.20221 (0.00096)
0.25	0.27117 (0.02656)	0.26065 (0.01271)	0.25163 (0.00464)	0.25127 (0.00247)	0.250819 (0.00163)	0.25057 (0.00126)
0.30	0.32079 (0.03176)	0.30376 (0.01637)	0.30416 (0.00585)	0.30308 (0.00330)	0.29970 (0.00182)	0.30071 (0.00165)
0.35	0.37739 (0.03911)	0.35926 (0.01770)	0.35122 (0.00781)	0.35287 (0.00377)	0.35328 (0.00248)	0.35211 (0.00180)

Table 7: Mean estimates and MSE of L_q using UMVUE

n	10	20	50	100	150	200
L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.15	0.15354 (0.01268)	0.14840 (0.00605)	0.15065 (0.00246)	0.15059 (0.00123)	0.14949 (0.00083)	0.15003 (0.00062)
0.20	0.20152 (0.01893)	0.19551 (0.00824)	0.19853 (0.00327)	0.19971 (0.00192)	0.19940 (0.00117)	0.20075 (0.00085)
0.25	0.25594 (0.02470)	0.25110 (0.01175)	0.25507 (0.00574)	0.25046 (0.00254)	0.24990 (0.00159)	0.25038 (0.00127)
0.30	0.29784 (0.03123)	0.29949 (0.01454)	0.30212 (0.00675)	0.30023 (0.00318)	0.29936 (0.00182)	0.30079 (0.00157)
0.35	0.34379 (0.03466)	0.35909 (0.02009)	0.35307 (0.00798)	0.34754 (0.00383)	0.35247 (0.00258)	0.35008 (0.00202)

Table 8: Mean estimates and MSE of L_q for Bayesian estimation under gamma prior ($\alpha = 2$, $\beta = 3$)

	n	10	20	50	100	150	200
Loss function	L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.15	0.15012 (0.00562)	0.15274 (0.00415)	0.15650 (0.00232)	0.15265 (0.00124)	0.15156 (0.00082)	0.15192 (0.00062)
	0.20	0.16943 (0.00585)	0.18549 (0.00449)	0.20036 (0.00280)	0.20023 (0.00168)	0.20031 (0.00118)	0.20176 (0.00090)
	0.25	0.18674 (0.00585)	0.21008 (0.00444)	0.23537 (0.00286)	0.24592 (0.00184)	0.24757 (0.00139)	0.24836 (0.00111)
	0.30	0.20790 (0.00573)	0.23232 (0.00433)	0.26303 (0.00262)	0.27930 (0.00171)	0.28662 (0.00130)	0.29063 (0.00106)
	0.35	0.21969 (0.00557)	0.25158 (0.00403)	0.28757 (0.00221)	0.30684 (0.00132)	0.31542 (0.00097)	0.32136 (0.00077)
PLF	0.15	0.16863 (0.03702)	0.16609 (0.02668)	0.16373 (0.01445)	0.15659 (0.00787)	0.15424 (0.00534)	0.15394 (0.00404)
	0.20	0.18690 (0.03494)	0.19783 (0.02467)	0.20737 (0.01401)	0.20439 (0.00831)	0.20321 (0.00581)	0.20398 (0.00442)
	0.25	0.20291 (0.03234)	0.22110 (0.02204)	0.24161 (0.01248)	0.24973 (0.00763)	0.25040 (0.00565)	0.25059 (0.00447)
	0.30	0.22234 (0.02888)	0.24209 (0.01955)	0.26822 (0.01039)	0.28245 (0.00630)	0.28894 (0.00464)	0.29249 (0.00372)
	0.35	0.23306 (0.02674)	0.25999 (0.01682)	0.29158 (0.00803)	0.30907 (0.00446)	0.31702 (0.00318)	0.32260 (0.00247)

Table 9: Mean estimates and MSE of L_q for for E-Bayesian estimation with $c = 1.5$

	n	10	20	50	100	150	200
Loss function	L_q	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)
SELF	0.15	* (*)	0.15168 (0.00431)	0.15462 (0.00237)	0.15369 (0.00128)	0.15250 (0.00085)	0.15225 (0.00063)
	0.20	* (*)	0.18447 (0.00467)	0.19707 (0.00286)	0.20082 (0.00171)	0.20237 (0.00120)	0.19985 (0.00090)
	0.25	0.18572 (0.00615)	0.20950 (0.00469)	0.23735 (0.00291)	0.24716 (0.00187)	0.24914 (0.00140)	0.24868 (0.00112)
	0.30	0.20777 (0.00595)	0.23328 (0.00449)	0.26363 (0.00264)	0.28154 (0.00170)	0.28633 (0.00130)	0.29124 (0.00107)
	0.35	0.21835(0.00583)	0.25190 (0.00409)	0.28692 (0.00223)	0.30860 (0.00130)	0.31634 (0.00096)	0.32256 (0.00076)
PLF	0.15	* (*)	0.16575 (0.02854)	0.16206 (0.01489)	0.15773 (0.00808)	0.15522 (0.00543)	0.15429 (0.00409)
	0.20	* (*)	0.19743 (0.02592)	0.20434 (0.01455)	0.20504 (0.00844)	0.20531 (0.00531)	0.20209 (0.00447)
	0.25	0.20301 (0.03459)	0.22118 (0.02336)	0.24365 (0.01262)	0.25101 (0.00771)	0.25198 (0.00568)	0.25094 (0.00452)
	0.30	0.22296 (0.03039)	0.24342 (0.02027)	0.26887 (0.01048)	0.28466 (0.00624)	0.28866 (0.00465)	0.29311 (0.00374)
	0.35	0.23257 (0.02845)	0.26053 (0.01726)	0.29100 (0.00812)	0.31078 (0.00436)	0.31790 (0.00312)	0.32374 (0.00243)

These values are not available

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Table 10: Mean estimates and MSE of L_q for H-Bayesian estimation for $c = 1.5$

	n	10	20	50	100	150	200
Loss function	L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.15	* (*)	0.15223 (0.00440)	0.15780 (0.00243)	0.15193 (0.00125)	0.15226 (0.00084)	0.15171 (0.00063)
	0.20	0.17213 (0.00607)	0.18414 (0.00470)	0.19725 (0.00283)	0.20692 (0.00172)	0.20120 (0.00120)	0.20287 (0.00092)
	0.25	0.18805 (0.00604)	0.21685 (0.00462)	0.23634 (0.00288)	0.24798 (0.00188)	0.24858 (0.00140)	0.25023 (0.00112)
	0.30	0.20874 (0.00600)	0.23663 (0.00445)	0.26669 (0.00261)	0.28249 (0.00170)	0.28869 (0.00129)	0.29230 (0.00106)
	0.35	0.22382 (0.00573)	0.25281 (0.00408)	0.28688 (0.00222)	0.30699 (0.00013)	0.31655 (0.00097)	0.32237 (0.00077)
PLF	0.15	* (*)	0.16648 (0.02851)	0.16546 (0.01493)	0.15594 (0.00803)	0.15498 (0.00543)	0.15375 (0.00408)
	0.20	0.19047 (0.03668)	0.19721 (0.02613)	0.20446 (0.01442)	0.20692 (0.00845)	0.20414 (0.00588)	0.20511 (0.00448)
	0.25	0.20506 (0.03402)	0.22807 (0.02243)	0.24263 (0.01259)	0.25183 (0.00770)	0.25143 (0.00569)	0.25248 (0.00450)
	0.30	0.22399 (0.03051)	0.24655 (0.01984)	0.27182 (0.01026)	0.28560 (0.00621)	0.29099 (0.00460)	0.29416 (0.00371)
	0.35	0.23752 (0.02740)	0.26138 (0.01714)	0.29096 (0.00816)	0.30924 (0.00450)	0.31812 (0.00314)	0.32359 (0.00243)

* These values are not available

Table 11: Bayesian Estimation for ρ under gamma prior

ρ	n	Mean (MSE)	Acc-rate	ESS	Geweke-Z score	Convergence
0.3	100	0.32030 (0.00342)	0.90669	275.579	1.25151	convergent
	200	0.28620 (0.00148)	0.88039	450.013	0.56360	convergent
	500	0.29411 (0.00064)	0.81399	937.766	0.67978	convergent
0.5	100	0.50432 (0.00481)	0.88274	399.346	0.73252	convergent
	200	0.52688 (0.00313)	0.84184	744.796	1.08707	convergent
	500	0.48756 (0.00116)	0.76443	1206.006	0.30743	convergent
0.7	100	0.65974 (0.00795)	0.86884	506.251	1.11855	convergent
	200	0.68492 (0.00341)	0.81844	890.420	0.57788	convergent
	500	0.68545 (0.00167)	0.72657	1411.824	0.20919	convergent

Table 12: H-Estimation for ρ under gamma prior

ρ	n	Mean (MSE)	Acc-rate	ESS	Geweke-Z score	Convergence
0.3	100	0.31264 (0.00323)	0.90800	259.305	1.21991	convergent
	200	0.28191 (0.00158)	0.88294	451.090	0.44234	convergent
	500	0.29411 (0.00064)	0.81399	937.766	0.67978	convergent
0.5	100	0.50158 (0.00493)	0.88589	389.404	0.72554	convergent
	200	0.52613 (0.00311)	0.84239	899.376	0.69573	convergent
	500	0.48691 (0.00118)	0.76658	1180.406	0.24362	convergent
0.7	100	0.65956 (0.00806)	0.86859	502.611	1.19893	convergent
	200	0.68609 (0.00339)	0.81874	899.398	0.77673	convergent
	500	0.68551 (0.00169)	0.72793	1403.149	0.27330	convergent

Table 13: Posterior predictive probabilities under Bayesian and H-Bayesian estimation

ρ	m	Bayesian	H-Bayesian	ρ	m	Bayesian	H-Bayesian
0.6	0	0.53365	0.52332	0.8	0	0.48935	0.48768
	1	0.33353	0.33721		1	0.34806	0.34590
	2	0.10583	0.11030		2	0.12544	0.12800
	3	0.02272	0.02441		3	0.03053	0.03166
	4	0.00371	0.00411		4	0.00565	0.00595
	5	0.00049	0.00056		5	0.00084	0.00090
	6	0.00006	0.00006		6	0.00011	0.00012
	≥ 7	0.00000	0.00000		7	0.00001	0.00001
					≥ 8	0.00000	0.00000

Table 14: Posterior predictive probabilities under Bayesian estimation using SIR method

ρ	m	$\mathcal{B}(1,2)$	$\mathcal{B}(2,5)$	$\mathcal{B}(5,2)$	$\mathcal{B}(7,3)$
0.6	0	0.54772	0.54775	0.54775	0.54778
	1	0.32972	0.32971	0.32971	0.32970
	2	0.09925	0.09923	0.09923	0.09922
	3	0.01992	0.01991	0.01991	0.01990
	4	0.00300	0.00300	0.00300	0.00300
	5	0.00036	0.00036	0.00036	0.00036
	6	0.00004	0.00004	0.00004	0.00004
	≥ 7	0.00000	0.00000	0.00000	0.00000
0.8	0	0.46524	0.46777	0.46518	0.46516
	1	0.35600	0.35531	0.35602	0.35602
	2	0.13621	0.13503	0.13624	0.13625
	3	0.03474	0.03424	0.03475	0.03476
	4	0.00664	0.00652	0.00665	0.00665
	5	0.00102	0.00099	0.00102	0.00102
	6	0.00013	0.00013	0.00013	0.00013
	7	0.00001	0.00001	0.00001	0.00001
	≥ 8	0.00000	0.00000	0.00000	0.00000

Table 15: Posterior predictive probabilities under H-Bayesian estimation using SIR method

ρ	m	$\mathcal{B}(1,2)$	$\mathcal{B}(2,5)$	$\mathcal{B}(5,2)$	$\mathcal{B}(7,3)$
0.6	0	0.54889	0.54884	0.54889	0.54892
	1	0.32925	0.32927	0.32925	0.32944
	2	0.09875	0.09877	0.09875	0.09874
	3	0.01975	0.01975	0.01975	0.01974
	4	0.00296	0.00296	0.00296	0.00296
	5	0.00036	0.00036	0.00036	0.00036
	6	0.00004	0.00004	0.00004	0.00004
	≥ 7	0.00000	0.00000	0.00000	0.00000
0.8	0	0.46465	0.46737	0.46456	0.46456
	1	0.35613	0.35539	0.35615	0.35614
	2	0.13648	0.13522	0.13652	0.13653
	3	0.03487	0.03432	0.03489	0.03489
	4	0.00668	0.00654	0.00669	0.00669
	5	0.00102	0.00099	0.00102	0.00102
	6	0.00013	0.00013	0.00013	0.00013
	7	0.00001	0.00001	0.00001	0.00001
	≥ 8	0.00000	0.00000	0.00000	0.00000

Table 16: MLE and UMVUE for the real-life example

Estimator	$\hat{\rho}$ (SD)	$\hat{L}_q$ (SD)
MLE	0.93500 (1.02077)	0.32758 (6.56046)
UMVUE	0.93500 (1.02077)	0.32667 (1.39084)

Table 17: Bayesian, E-Bayesian and H-Bayesian estimation for the real-life example

Estimator	$\hat{\rho}_{SELF}$ (SD)	$\hat{\rho}_{PLF}$ (SD)	$\hat{L}_{qSELF}$ (SD)	$\hat{L}_{qPLF}$ (SD)
Bayesian	0.91156 (0.00277)	0.91307 (0.00303)	0.31402 (0.00097)	0.31556 (0.00309)
E-Bayesian	0.91313 (0.00274)	0.91463 (0.00300)	0.31495 (0.00096)	0.31647 (0.00305)
H-Bayesian	0.91343 (0.00271)	0.91492 (0.00296)	0.31513 (0.00094)	0.31662 (0.00299)